

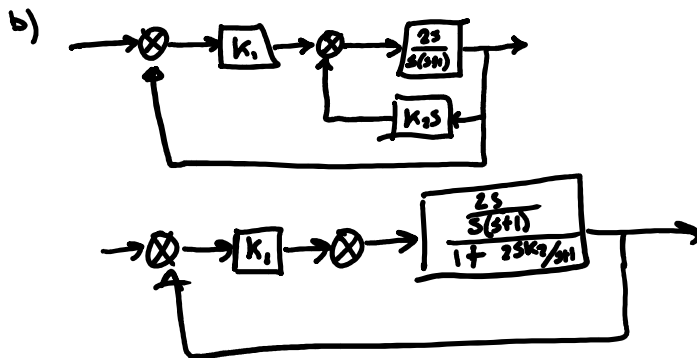
5.72, 6.1, 9, 12, 21, 23, 7.5, 10

$$5.72) a) \frac{2s}{s(s+1)} = \frac{2s}{s^2+s+2s} = T(s)$$

$$\omega_n = 5 \quad 2s\omega_n = 1 \quad \zeta = \frac{1}{10}$$

$$\%OS = \exp\left[\frac{-3\pi}{\sqrt{1-\zeta^2}}\right] \times 100$$

$$= \exp\left[\frac{-\pi/10}{\sqrt{99/100}}\right] \times 100 = \boxed{72.9\%}$$



$$T(s) = \frac{2sK_1}{s(s+1) + 2sK_2s}$$

$$1 + \frac{2sK_1}{s(s+1) + 2sK_2s}$$

$$T(s) = \frac{2sK_1}{s^2 + s + 2sK_2s + 2sK_1}$$

$$T(s) = \frac{2sK_1}{s^2 + (1+2sK_2)s + 2sK_1}$$

$$\omega_n = 5\sqrt{K_1}$$

$$2s\omega_n = (1+2sK_2)$$

$$\zeta = \frac{1+2sK_2}{10\sqrt{K_1}}$$

$$T_s = \frac{4}{s\omega_n} = \frac{8}{(1+2sK_2)} = 0.25$$

$$\frac{31}{25} = K_2$$

$$\%OS = \exp\left[\frac{-32/10\sqrt{K_1}\pi}{\sqrt{1 - \frac{32^2}{100K_1}}}\right] \times 100$$

$$0.2 = \exp\left[\frac{-32}{\sqrt{K_1}}\pi}{\sqrt{100 - \frac{32^2}{K_1}}}\right]$$

... 32

$$m(0.2) = -\frac{\sqrt{k_1}}{\sqrt{100 - \frac{32^2}{k_1}}}$$

$$\ln(0.2) \sqrt{100 - \frac{32^2}{k_1}} = -\frac{32}{\sqrt{k_1}} \pi$$

$k_1 = 49.26$

$k_2 = 1.24$

6.1) $s^5 + 3s^4 + 5s^3 + 4s^2 + s + 3$

Roots are:

$$s = -1.63, -0.94 \pm 1.5i, 0.26 \pm 0.72i$$

the system is unstable because it has roots in the right half of the plane

6.9) $T(s) = \frac{240}{(s+1)(s+2)(s+3)(s+4) + 240}$

$$T(s) = \frac{240}{(s^2 + 3s + 2)(s+3)(s+4) + 240}$$

$$T(s) = \frac{240}{(s^3 + 6s^2 + 11s + 6)(s+4) + 240}$$

$$T(s) = \frac{240}{s^4 + 10s^3 + 35s^2 + 50s + 264}$$

the transfer function has two poles in the right hand plane and is unstable

6.12) $G(s) = \frac{K(s+1)}{s(s-2)(s+3)}$

$$G_{cl}(s) = \frac{K(s+1)}{s(s-2)(s+3) + K(s+1)}$$

$$= \frac{Ks + K}{s^3 + s^2 + (K-6)s + K}$$

$$\begin{array}{ccc} s^3 & 1 & 1 & 0 \\ s^2 & K-6 & K & 0 \end{array}$$

$$s^1 - \frac{\begin{vmatrix} 1 & 1 \\ K-6 & K \end{vmatrix}}{K-6} = \frac{-6}{K-6} \quad -\frac{\begin{vmatrix} 1 & 0 \\ K-6 & 0 \end{vmatrix}}{K-6} = 0 \quad -\frac{\begin{vmatrix} 1 & 0 \\ K-6 & 0 \end{vmatrix}}{K-6} = 0$$

$$K-6 > 0; K < 6$$

$$-\frac{6}{K-6} > 0; \frac{1}{K-6} < 0$$

$-6 < K < 6$

$$s^0 - \frac{\begin{vmatrix} K-6 & K-6 \\ 6 & 0 \end{vmatrix}}{K-6} = 6 \quad \frac{6}{K-6} > 0 \quad 6K+36 > 0 \quad K > -6$$

6.21) $G_{cl}(s) = \frac{k(s+6)}{s^3 + 5s^2 + (4+k)s + 6k}$

$$s^3 + 5s^2 + 4s + ks + k6$$

$$\begin{array}{ccc} 1 & 4+k & 0 \\ 5 & 6k & 0 \end{array}$$

$$-\left| \begin{array}{cc} 1 & 4+k \\ 5 & 6k \end{array} \right| = \frac{20-k}{5} \quad 0 \quad 0 \quad \frac{20-k}{5} > 0; k < 20$$

$$-\left| \begin{array}{cc} 5 & 6k \\ 20-k & 0 \end{array} \right| = 6k \quad 0 \quad 0 \quad \begin{array}{l} 6k > 0 \\ k > 0 \end{array}$$

$$\boxed{0 < k < 20}$$

$$6.23) G(s) = \frac{k(s+3)(s+5)}{(s-2)(s-4) + k(s+3)(s+5)}$$

$$G(s) = \frac{k(s+3)(s+5)}{(k+1)s^2 + (8k-6)s + (3+15k)} \quad \begin{array}{ccc} s^2 & k+1 & 3+15k \\ s^1 & 8k-6 & 0 \\ s^0 & 3k-6 & 0 \end{array} \quad \begin{array}{l} -15k-8 < 0 \\ -15k-8 > 0 \end{array}$$

$$\boxed{k < -\frac{8}{15}}$$

7.5) step input $-20\mu(t)$

$$G(s) = \frac{500}{(s+2)(s^2+8s+12)}$$

$$\lim_{s \rightarrow 0} G(s) = \frac{500}{2 \cdot 12} = \frac{125}{84}$$

$$G(s) = \frac{20}{1 + \frac{125}{84}} = \frac{1680}{209} = \boxed{8.03}$$

7.10)

$$G(s) = \frac{5000}{s^2 + 75s} \quad 2\zeta\omega_n = 75 \Rightarrow \zeta = \frac{75}{100\sqrt{2}}$$

$$a) \%OS = \exp\left[-\frac{75\pi}{100\sqrt{2}}\right] \times 100 = \boxed{14.01\%}$$

$$b) T_s = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{75}{100\sqrt{2}} \cdot 80\sqrt{2}} = \frac{4}{\frac{75}{2}} = \frac{8}{75} = \boxed{0.107s}$$

$$c) \lim_{s \rightarrow 0} \frac{5}{1 + \frac{5000}{s^2 + 75s}} = \frac{5(s^2 + 75s)}{s^2 + 75s + 5000} = \frac{0}{5000} = \boxed{0}$$